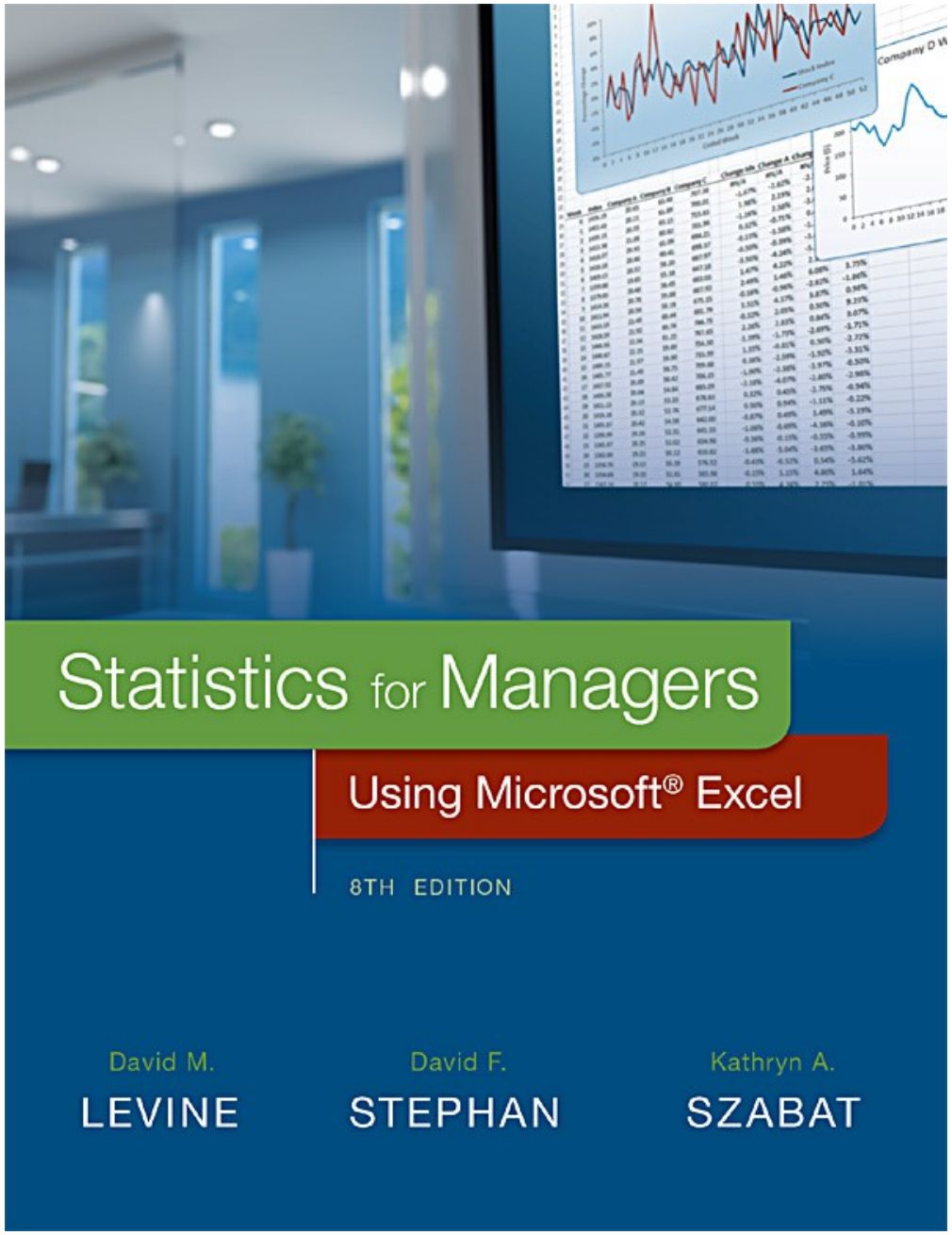


Chapter 9

Fundamentals of Hypothesis Testing: One-Sample Tests

Paul Andrew Gorgonio

Objectives

In this chapter, you learn:

- The basic principles of hypothesis testing.
- How to use hypothesis testing to test a mean or proportion.
- To identify the assumptions of each hypothesis-testing procedure, how to evaluate them, and the consequences if they are seriously violated.
- The pitfalls & ethical issues involved in hypothesis. testing.
- How to avoid the pitfalls involved in hypothesis testing.

What is a Hypothesis?

- A hypothesis is a claim (assertion) about a population parameter:
  - population mean:

    Example: The mean monthly cell phone bill in this city is $\mu = \$42$.

  - population proportion:

    Example: The proportion of adults in this city with cell phones is $\pi = 0.88$.



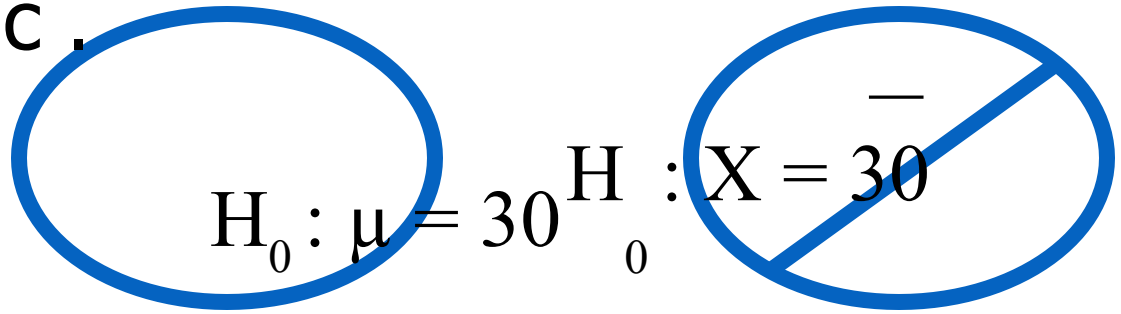
The Null Hypothesis, H_0

- States the claim or assertion to be tested.

Example: The mean diameter of a manufactured bolt is 30mm.

$(H_0 : \mu = 30)$

- Is always about a population parameter, not about a sample statistic .



The Null Hypothesis, H_0

- Begin with the assumption that the null hypothesis is true.
  - Similar to the notion of innocent until proven guilty.
- Represents the current belief in a situation.
- Always contains "=", or "≤", or "≥" sign.
- May or may not be rejected.

The Alternative Hypothesis, H_1

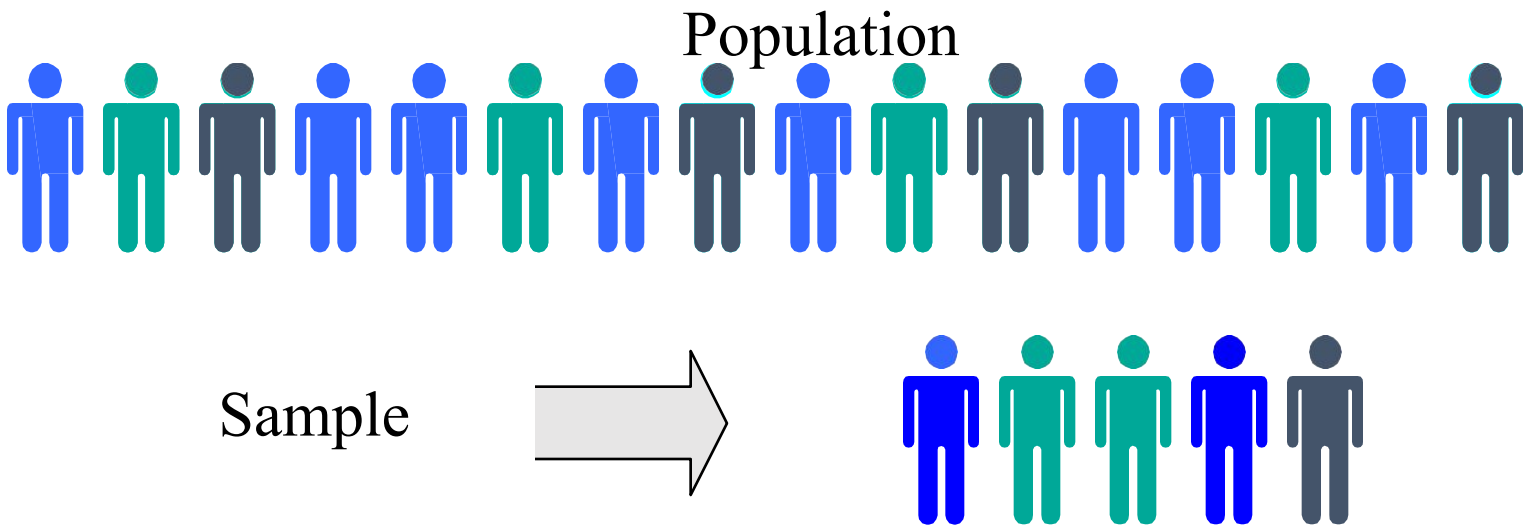
- Is the opposite of the null hypothesis.
    - e.g., The mean diameter of a manufactured bolt is not equal to 30mm (H_1: $\mu \neq 30$).
- Challenges the status quo.
- Never contains the "=", or "≤", or "≥" sign.
- May or may not be proven.
- Is generally the hypothesis that the researcher is trying to prove.

The Hypothesis Testing Process

Claim: The population mean age is 50.

$H_0: \mu = 50, H_1: \mu \neq 50$

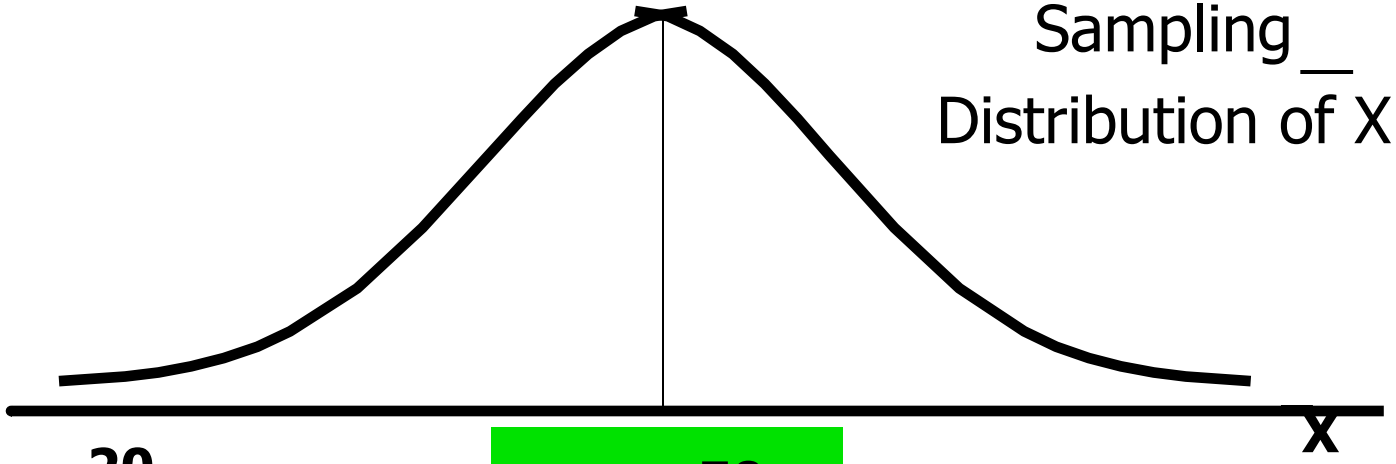
Sample the population and find the sample mean



The Hypothesis Testing Process

- Suppose the sample mean age was $\bar{X} = 20$.
- This is significantly lower than the claimed mean population age of 50.
- If the null hypothesis were true, the probability of getting such a different sample mean would be very small, so you reject the null hypothesis .
- In other words, getting a sample mean of 20 is so unlikely if the population mean was 50, you conclude that the population mean must not be 50.

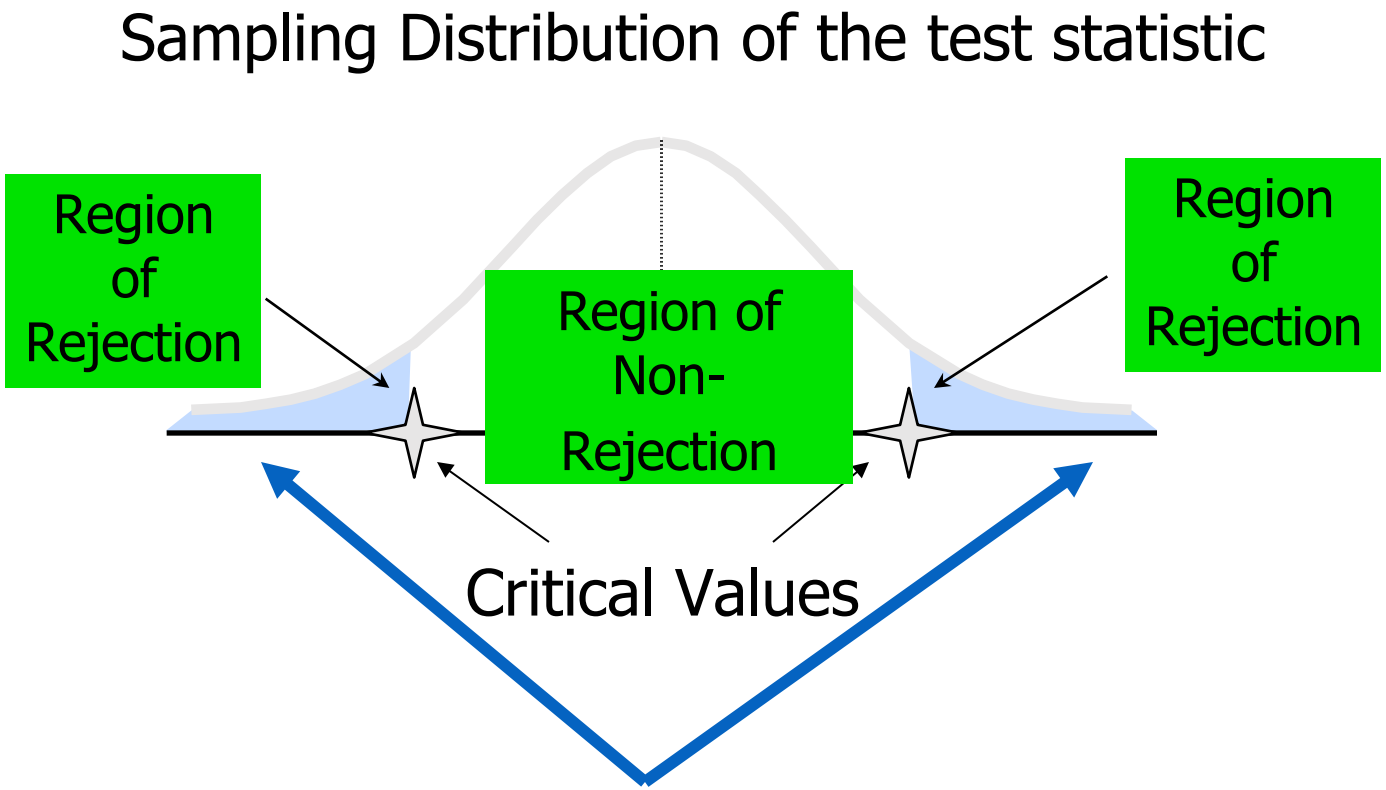
The Hypothesis Testing Process



The Test Statistic and Critical Values

- If the sample mean is close to the stated population mean, the null hypothesis is not rejected.
- If the sample mean is far from the stated population mean, the null hypothesis is rejected.
- How far is "far enough" to reject H_0?
- The critical value of a test statistic creates a "line in the sand" for decision making -- it answers the question of how far is far enough.

The Test Statistic and Critical Values



Risks in Decision Making Using Hypothesis Testing

TYPE I ERROR	Type II ERROR
• Reject a true null hypothesis. • A type I error is a "false alarm." • The probability of a Type I Error is α. • Called level of significance of the test. • Set by researcher in advance.	• Failure to reject a false null hypothesis. • Type II error represents a "missed opportunity." • The probability of a Type II Error is **β.**

Possible Errors in Hypothesis Test Decision Making

Possible Hypothesis Test Outcomes		
	Actual Situation	
Decision	H_0 True	H_0 False
Do Not Reject H_0	Correct Decision Confidence = 1 - α	Type II Error P(Type II Error) = β
Reject H_0	Type I Error P(Type I Error) = α	Correct Decision Power = (1 – β)

Possible Errors in Hypothesis Test Decision Making

- The confidence coefficient (1-α) is the probability of not rejecting H_0 when it is true.
- The confidence level of a hypothesis test is (1-α)*100%.
- The power of a statistical test (1-β) is the probability of rejecting H_0 when it is false.

Type I & II Error Relationship

- Type I and Type II errors cannot happen at the same time.
  - A Type I error can only occur if H_0 is true.
  - A Type II error can only occur if H_0 is false.

If Type I error probability (α) increases, then Type II error probability (β) decreases.

Factors Affecting Type II Error

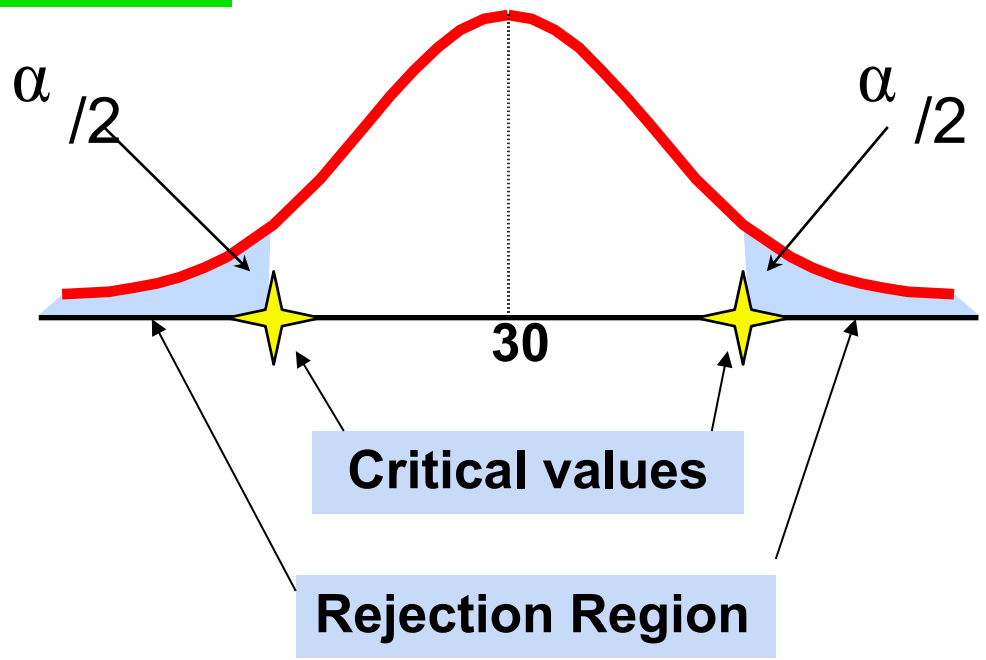
- All else equal,
  - β ↑ when the difference between hypothesized parameter and its true value ↓
  - β ↑ when α ↓
  - β ↑ when σ ↑
  - β ↑ when n ↓

Level of Significance and the Rejection Region

H_0: $\mu = 30$

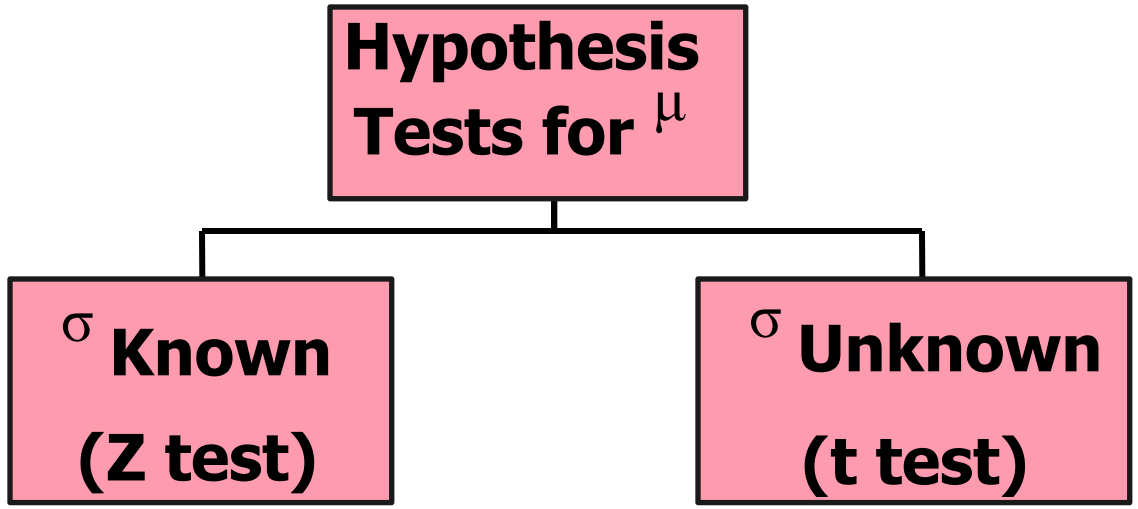
H_1: $\mu \neq 30$

Level of significance = α



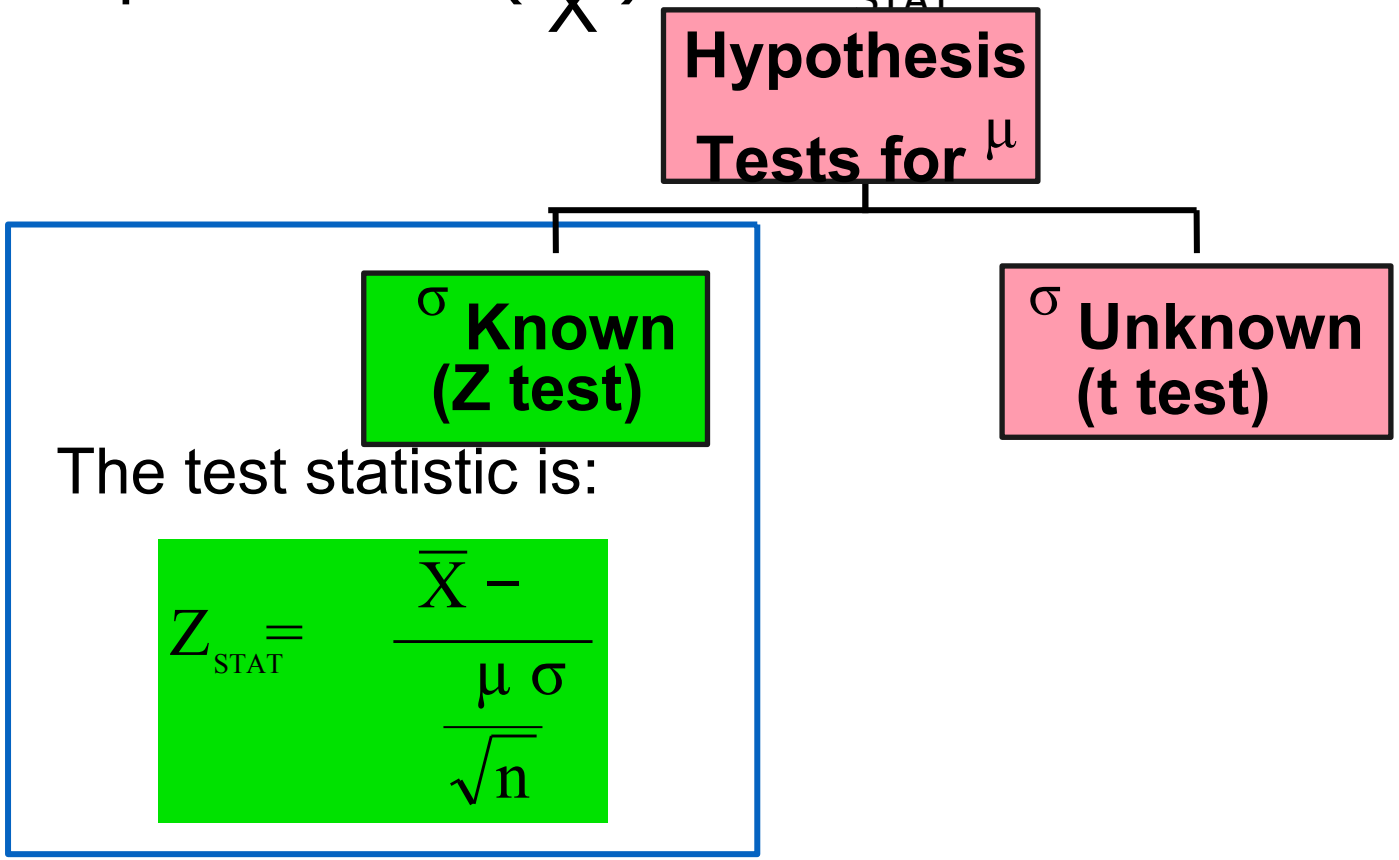
This is a two-tail test because there is a rejection region in both tails

Hypothesis Tests for the Mean



Z Test of Hypothesis for the Mean (σ Known)

- Convert sample statistic ($\bar{X}$) to a Z_{STAT} test statistic.

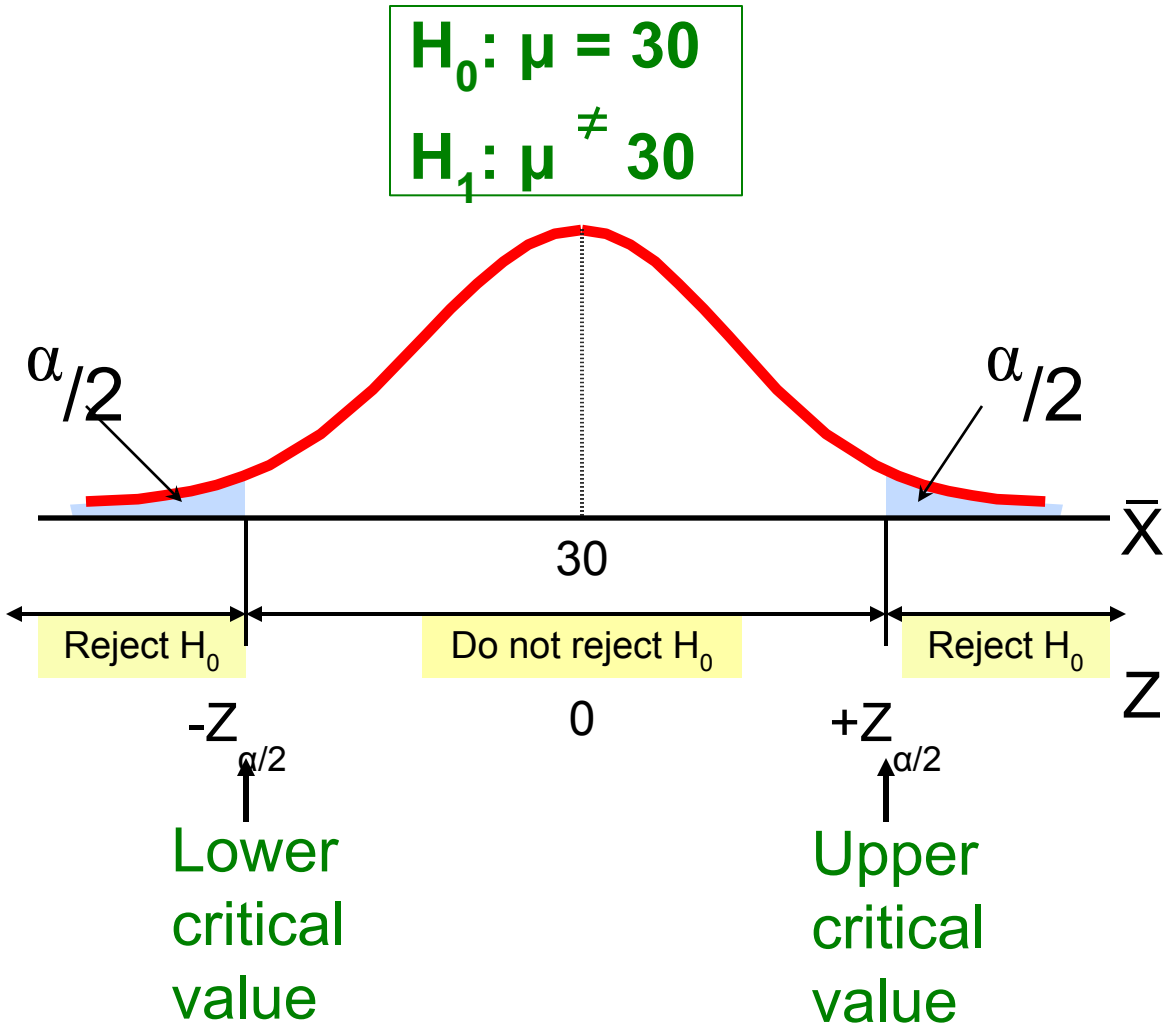


Critical Value Approach to Testing

- For a two-tail test for the mean, σ known:
- Convert sample statistic ($\bar{X}$) to test statistic (Z_{STAT}.)
- Determine the critical Z values for a specified level of significance α from a table or by using computer software.
- Decision Rule: If the test statistic falls in the rejection region, reject H_0 otherwise do not reject H_0.

Two-Tail Tests

- There are two cutoff values (critical values), defining the regions of rejection.



Steps in The Critical Value Approach To Hypothesis Testing

1. State the null hypothesis, H_0 and the alternative hypothesis, H_1.
2. Choose the level of significance, α, and the sample size, n. The level of significance is based on the relative importance of Type I and Type II errors in the situation.
3. Determine the appropriate test statistic and sampling distribution.
4. Determine the critical values that divide the rejection and non-rejection regions.

Steps in The Critical Value Approach To Hypothesis Testing

5. Collect the sample data, organize the results, and compute the value of the test statistic.
6. Make the statistical decision, determine whether the assumptions are valid, and state the managerial conclusion in the context of the theory, claim, or assertion being tested. If the test statistic falls into the non-rejection region, you do not reject the null hypothesis H_0. If the test statistic falls into the rejection region, reject the null hypothesis.

Hypothesis Testing Example:

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume $\sigma = 0.8$.

1. State the appropriate null and alternative hypotheses:
   - $H_0: \mu = 30$ $H_1: \mu \neq 30$ (This is a two-tail test).

2. Specify the desired level of significance and the sample size:
   - Suppose that $\alpha = 0.05$ and $n = 100$ are chosen for this test.

3. Determine the appropriate technique:
   - σ is assumed known so this is a Z test.

Hypothesis Testing Example:

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume σ = 0.8.

4. Determine the critical values:
   - For $\alpha = 0.05$ the critical Z values are ±1.96.

5. Collect the data and compute the test statistic.
   - Suppose the sample results are:

     n = 100, $\bar{X} = 29.84$ ($\sigma = 0.8$ is assumed known).

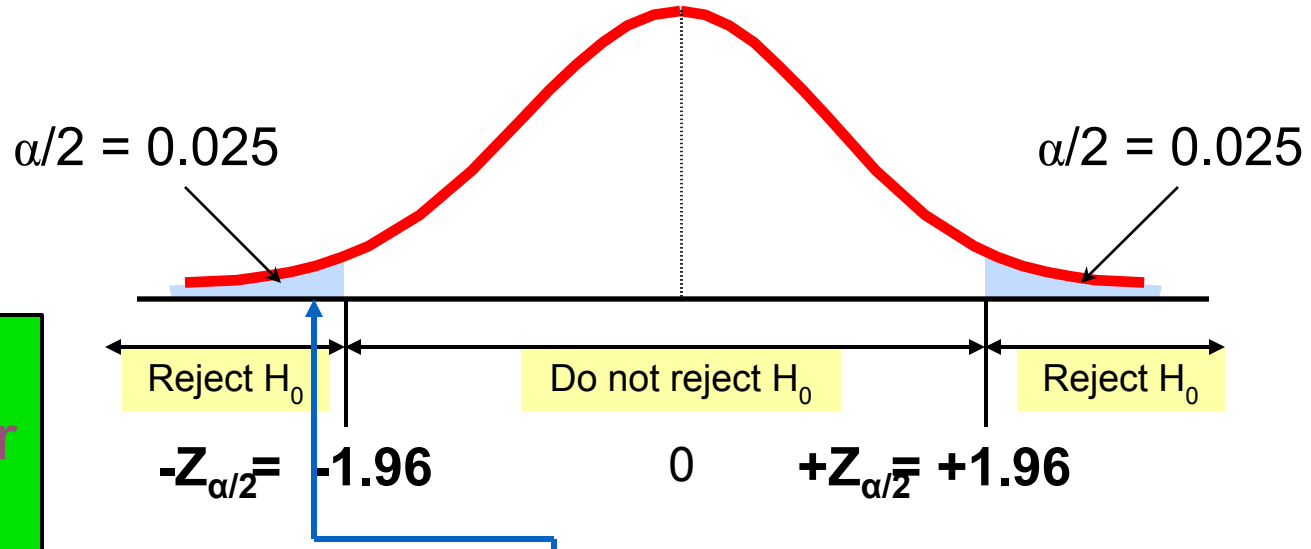
So the test statistic is:

$$Z_{STAT} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{29.84 - 30}{\frac{0.8}{\sqrt{100}}} = \frac{-0.16}{0.08} = -2.0$$

Hypothesis Testing Example:

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume σ = 0.8.

6. Is the test statistic in the rejection region?



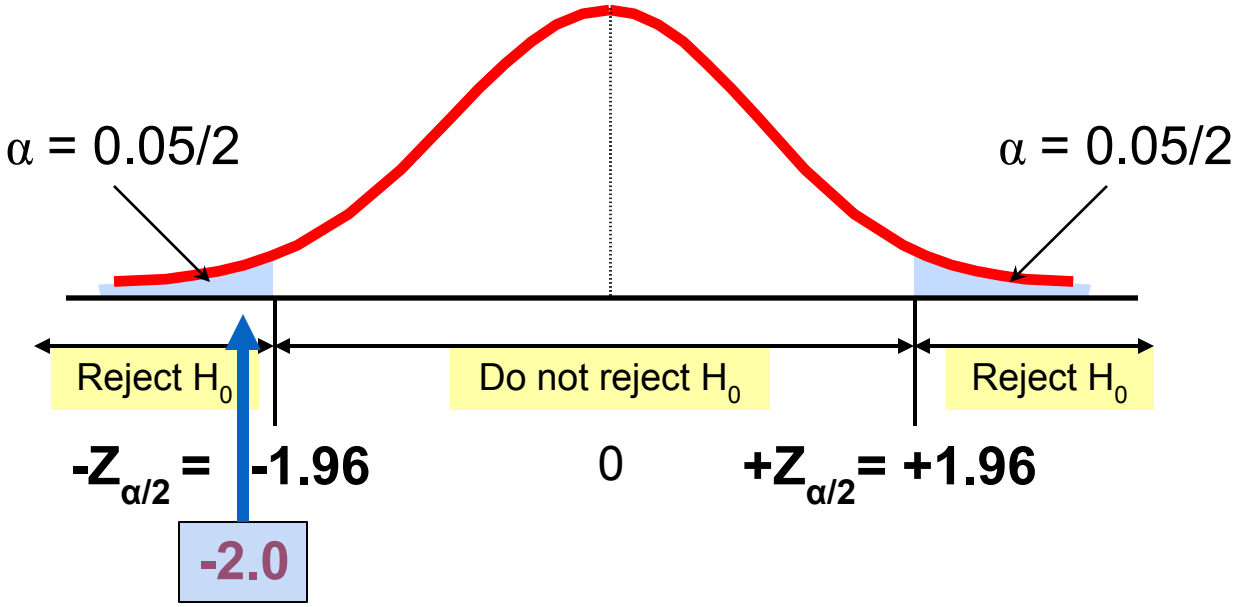
Reject H_0 if $Z_{STAT} < -1.96$ or $Z_{STAT} > 1.96$; otherwise do not reject H_0.

Here, $Z_{STAT} = -2.0 < -1.96$, so the test statistic is in the rejection region.

Hypothesis Testing Example:

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume σ = 0.8.

6. Reach a decision and interpret the result.



Since Z_{STAT} = -2.0 < -1.96, reject the null hypothesis and conclude there is sufficient evidence that the mean diameter of a manufactured bolt is not equal to 30.

p-Value Approach to Testing

- p-value: Probability of obtaining a test statistic equal to or more extreme than the observed sample value given H_0 is true.
  - The p-value is also called the observed level of significance.
  - It is the smallest value of α for which H_0 can be rejected.

p-Value Approach to Testing: Interpreting the p-value

- Compare the p-value with α:
  - If p-value $<$ α, reject H_0.
  - If p-value $\geq \alpha$, do not reject H_0.

- Remember: If the p-value is low, then H_0 must go.

The 5-Step p-Value Approach to Hypothesis Testing

1. State the null hypothesis, H_0 and the alternative hypothesis, H_1.
2. Choose the level of significance, α, and the sample size, n. The level of significance is based on the relative importance of the risks of a type I and a type II error.
3. Determine the appropriate test statistic and sampling distribution.
4. Collect the sample data, compute the value of the test statistic and the p-value.
5. Make the statistical decision and state the managerial conclusion in the context of the theory, claim, or assertion being tested. If the p-value is < α reject H_0.

p-value Hypothesis Testing Example

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume $\sigma = 0.8$.

1. State the appropriate null and alternative hypotheses:
   - $H_0: \mu = 30$ $H_1: \mu \neq 30$ (This is a two-tail test).

2. Specify the desired level of significance and the sample size:
   - Suppose that $\alpha = 0.05$ and $n = 100$ are chosen for this test.

p-value Hypothesis Testing Example

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume σ = 0.8.

3. Determine the appropriate technique:
   - σ is assumed known so this is a Z test.
4. Collect the data, compute the test statistic and the p-value.
   - Suppose the sample results are:

     n = 100, $\bar{X}$ = 29.84 (σ = 0.8 is assumed known.)

So the test statistic is:

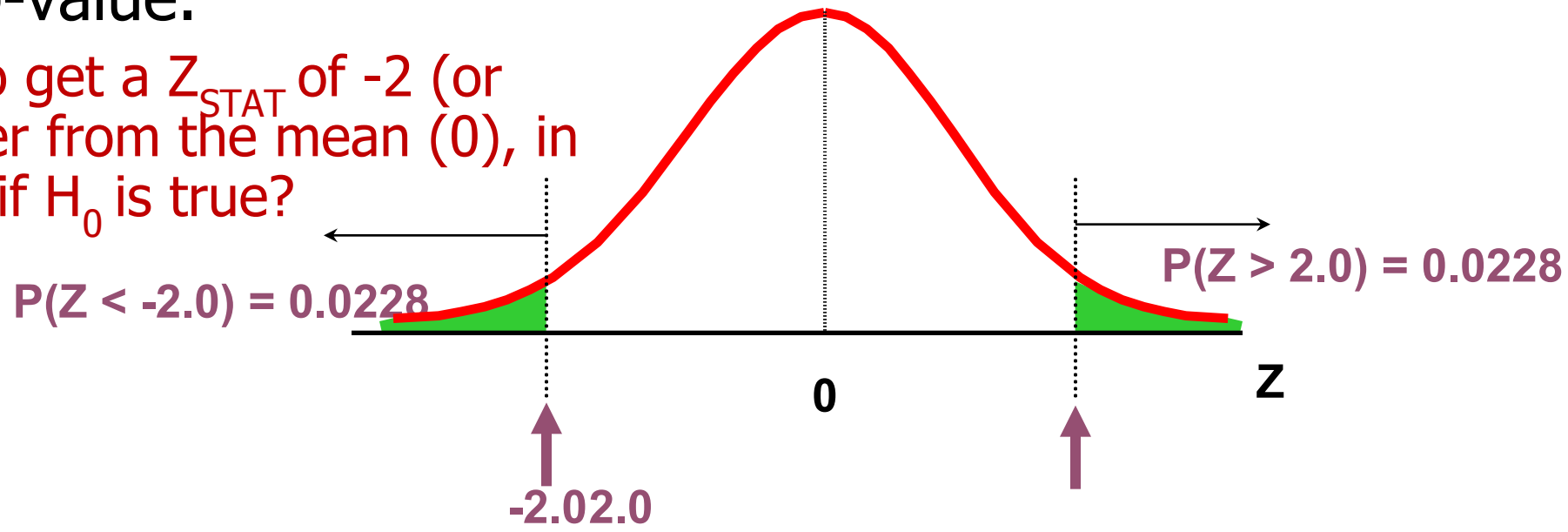
$$Z_{STAT} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{29.84 - 30}{\frac{0.8}{\sqrt{100}}} = \frac{-0.16}{0.08} = -2.0$$

p-value Hypothesis Testing Example

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume σ = 0.8.

4. Calculate the p-value.

How likely is it to get a Z_{STAT} of -2 (or something further from the mean (0), in either direction) if H_0 is true?



p-value = 0.0228 + 0.0228 = 0.0456

p-value Hypothesis Testing Example

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume σ = 0.8.

5. Is the p-value < α?

   Since p-value = 0.0456 < α = 0.05 Reject H_0.

   State the managerial conclusion in the context of the situation.

   There is sufficient evidence to conclude the mean diameter of a manufactured bolt is not equal to 30mm.

Connection Between Two Tail Tests and Confidence Intervals

- For $\bar{X} = 29.84$, $\sigma = 0.8$ and $n = 100$, the 95% confidence interval is:

$$29.84 - (1.96)\frac{0.8}{\sqrt{100}} \quad \text{to} \quad 29.84 + (1.96)\frac{0.8}{\sqrt{100}}$$

$$29.6832 \le \mu \le 29.9968$$

- Since this interval does not contain the hypothesized mean (30), we reject the null hypothesis at $\alpha = 0.05$.

Do You Ever Truly Know σ?

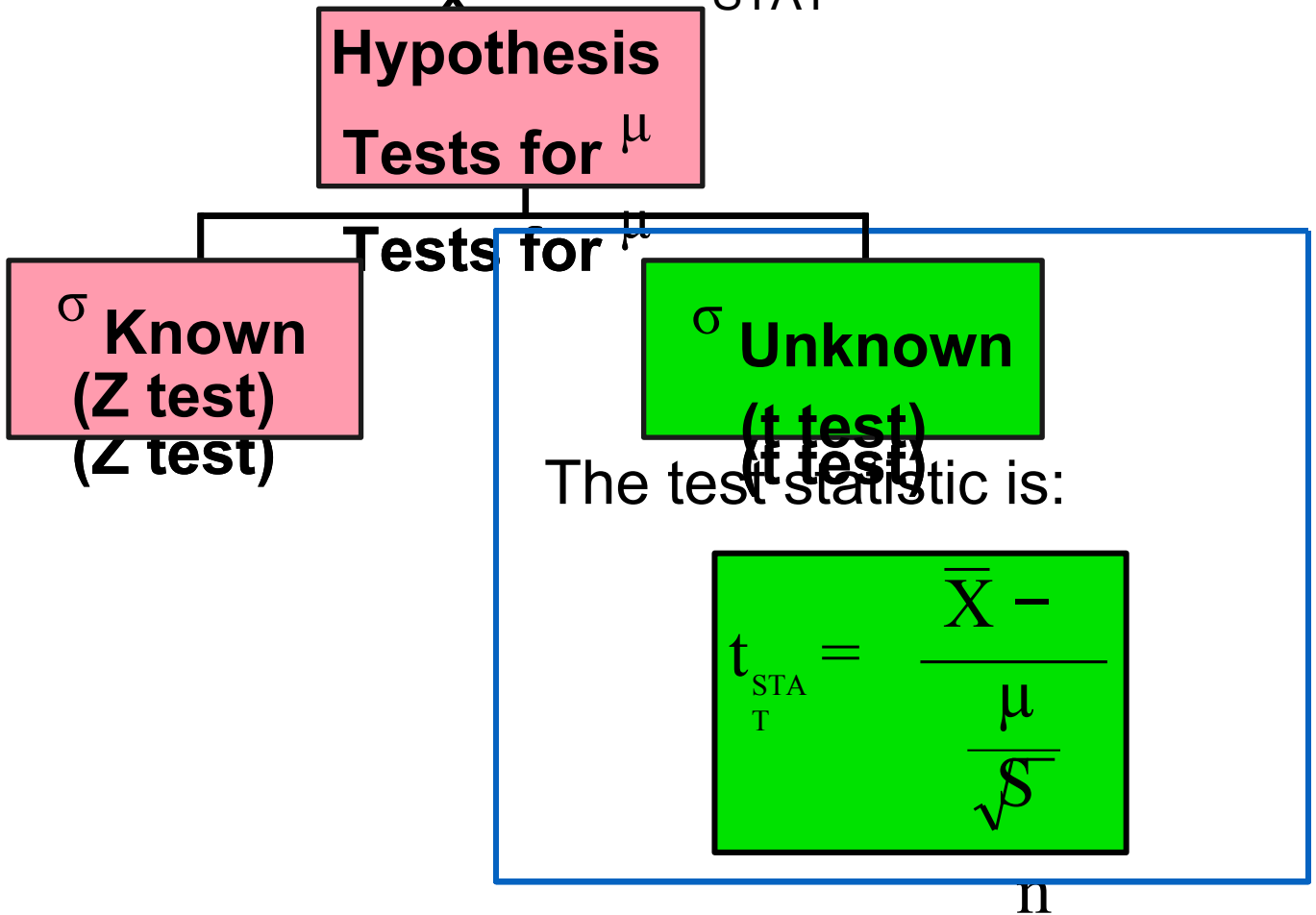
- Probably not!
- In virtually all real world business situations, σ is not known.
- If there is a situation where σ is known then μ is also known (since to calculate σ you need to know μ.)
- If you truly know μ there would be no need to gather a sample to estimate it.

Hypothesis Testing for the Mean: σ Unknown

- If the population standard deviation is unknown, you instead use the sample standard deviation S.
- Because of this change, you use the t distribution instead of the Z distribution to test the null hypothesis about the mean.
- When using the t distribution you must assume the population you are sampling from follows a normal distribution.
- All other steps, concepts, and conclusions are the same.

t Test of Hypothesis for the Mean (σ Unknown)

- Convert sample statistic ($\bar{X}$) to a t_{STAT} test statistic



Example: Two-Tail Test (σ Unknown)

The mean cost of a hotel room in New York is said to be $168 per night. To determine if this is true, a random sample of 25 hotels is taken and resulted in an X of $172.50 and an S of
$15.40. Test the appropriate hypotheses at $\alpha = 0.05$.

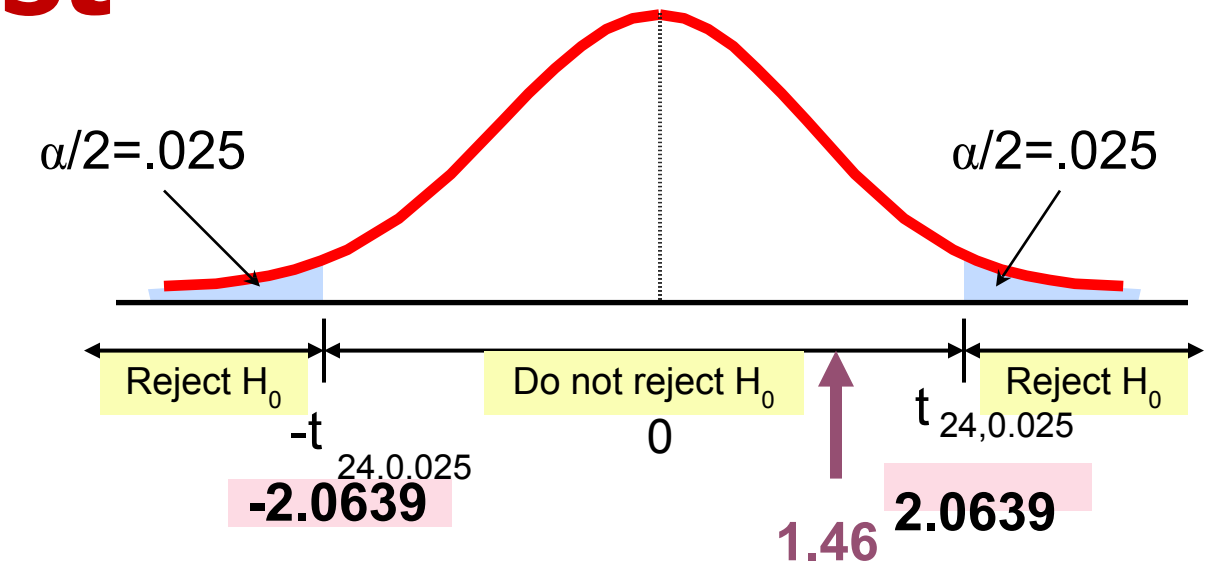
(Assume the population distribution is normal)

$$H_0: \mu = 168$$
$$H_1: \mu \neq 168$$

Example Solution: Two-Tail t Test

$H_0: \mu = 168$

$H_1: \mu \neq 168$



- $\alpha = 0.05$.
- n = 25, df = 25-1=24.
- σ is unknown, so use a t statistic.
- Critical Value: $\pm t_{24,0.025} = \pm 2.0639$.

$$tSTAT = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{172.50 - 168}{15.40/\sqrt{25}} = 1.46$$

Do not reject H_0: insufficient evidence that true mean cost is different from $168.

Example Two-Tail t Test Using A p-value from Excel

- Since this is a t-test we cannot calculate the p-value without some calculation aid.
- The Excel output below does this:

t Test for the Hypothesis of the Mean

Data		
Null Hypothesis μ=	$	168.00
Level of Significance		0.05
Sample Size		25
Sample Mean	$	172.50
Sample Standard Deviation	$	15.40

Intermediate Calculations			
Standard Error of the Mean	$	3.08	=B8/SQRT(B6)
Degrees of Freedom		24	=B6-1
t test statistic 1.46			=(B7-B4)/B11
Two-Tail Test			
Lower Critical Value	-2.0639		=-T.INV.2T(B5,B12)
Upper Critical Value	2.0639		=T.INV.2T(B5,B12)
p-value 0.157			=T.DIST.2T(ABS(B13),B12,2)
Do Not Reject Null Hypothesis			=IF(B18<B5, "Reject null hypothesis", "Do not reject null hypothesis")

p-value > α
So do not reject H_0

Connection of Two Tail Tests to Confidence Intervals

- For $\bar{X} = 172.5$, $S = 15.40$ and $n = 25$, the 95% confidence interval for μ is:

$$172.5 - (2.0639)\ 15.4/\sqrt{25} \quad \text{to} \quad 172.5 + (2.0639)\ 15.4/\sqrt{25}$$

$$166.14 \leq \mu \leq 178.86$$

- Since this interval contains the Hypothesized mean (168), we do not reject the null hypothesis at $\alpha = 0.05$.

One-Tail Tests

- In many cases, the alternative hypothesis focuses on a particular direction:

$H_0: \mu \geq 3$
$H_1: \mu < 3$

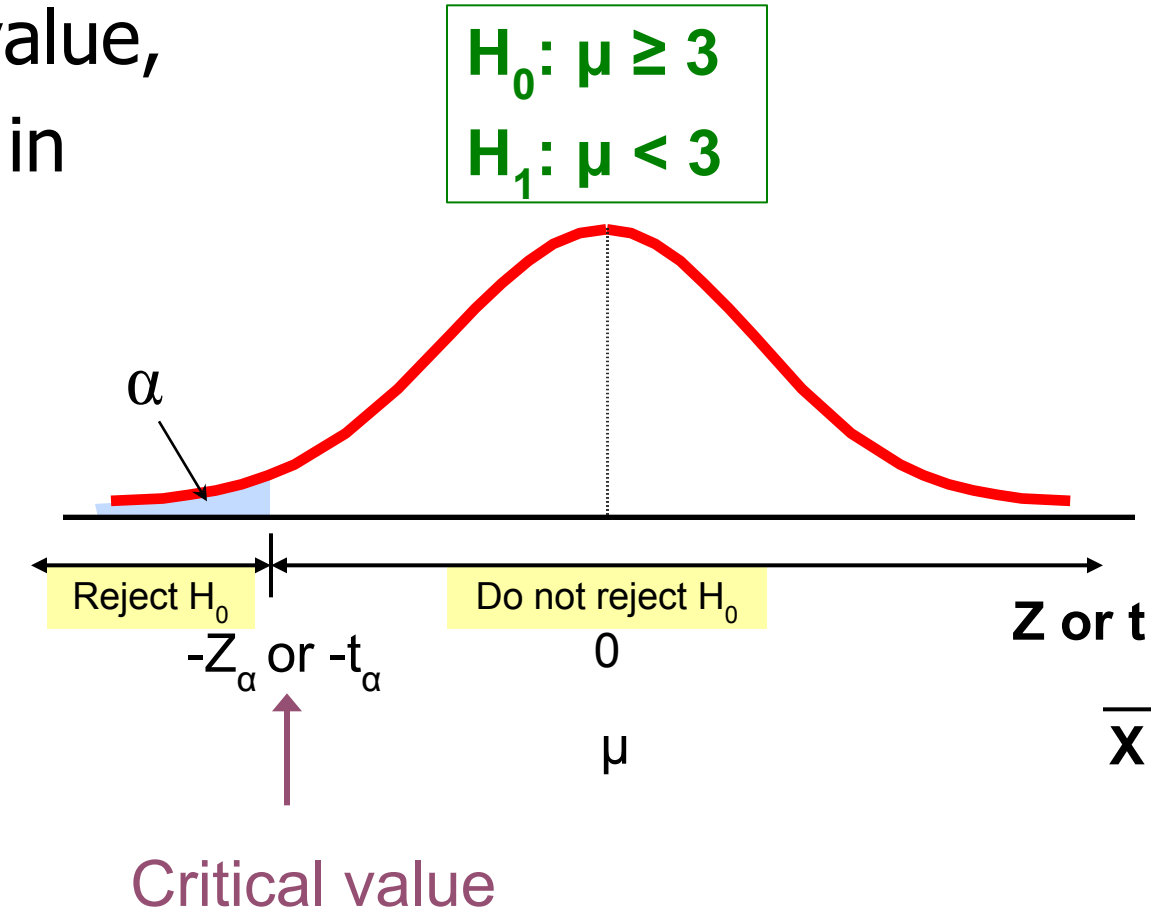
→ This is a lower-tail test since the alternative hypothesis is focused on the lower tail below the mean of 3.

$H_0: \mu \leq 3$
$H_1: \mu > 3$

→ This is an upper-tail test since the alternative hypothesis is focused on the upper tail above the mean of 3.

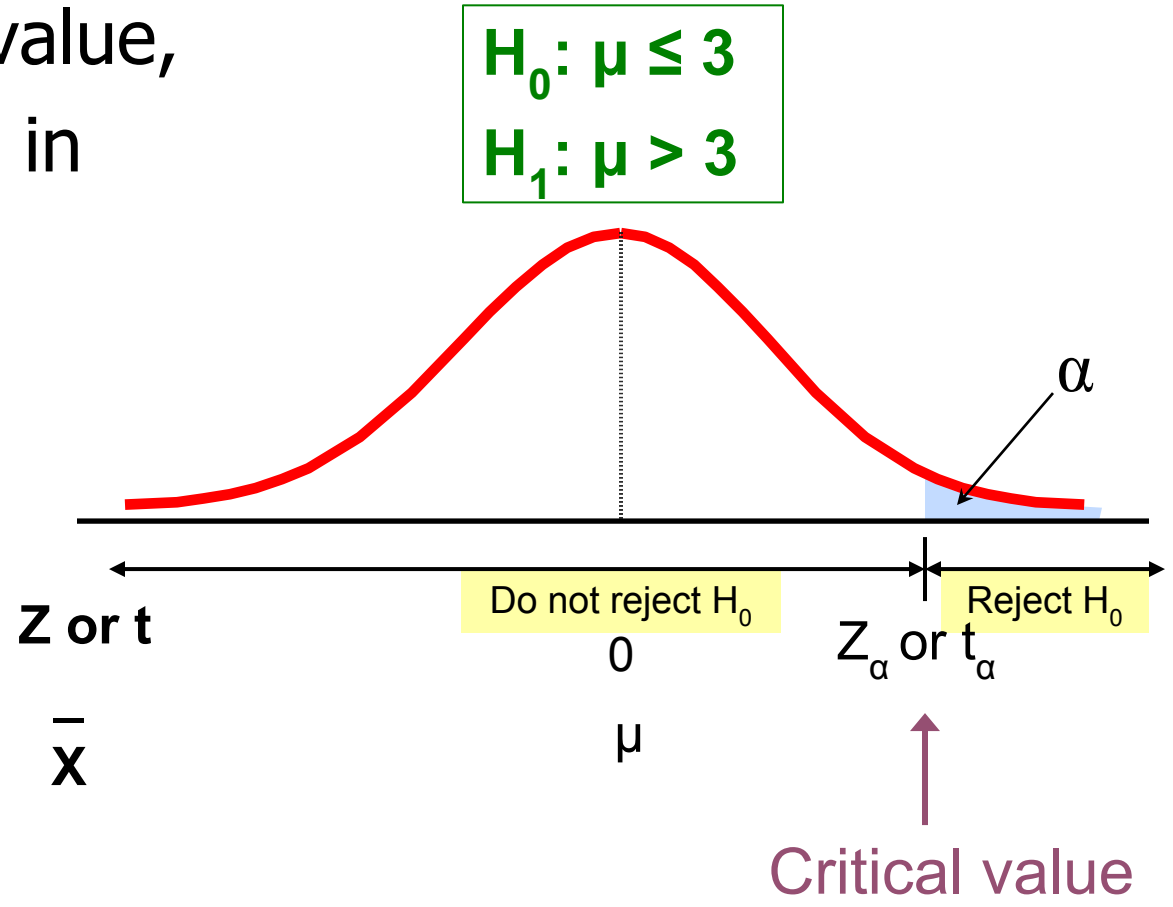
Lower-Tail Tests

- There is only one critical value, since the rejection area is in only one tail.



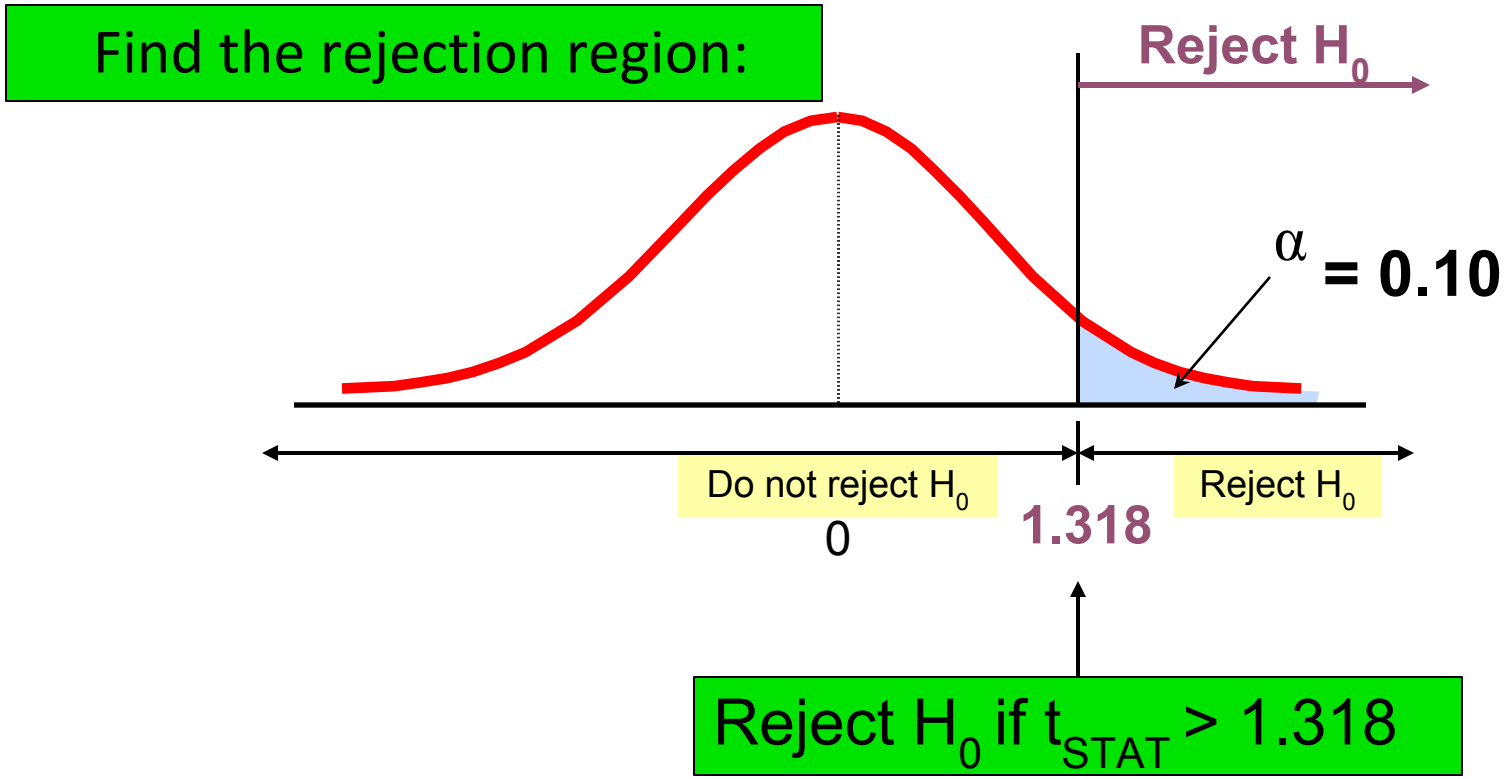
Upper-Tail Tests

- There is only one critical value, since the rejection area is in only one tail.



Example: Find Rejection Region

- Suppose that $\alpha = 0.10$ is chosen for this test and $n = 25$.



Example: Test Statistic

Obtain sample and compute the test statistic.

Suppose a sample is taken with the following results:

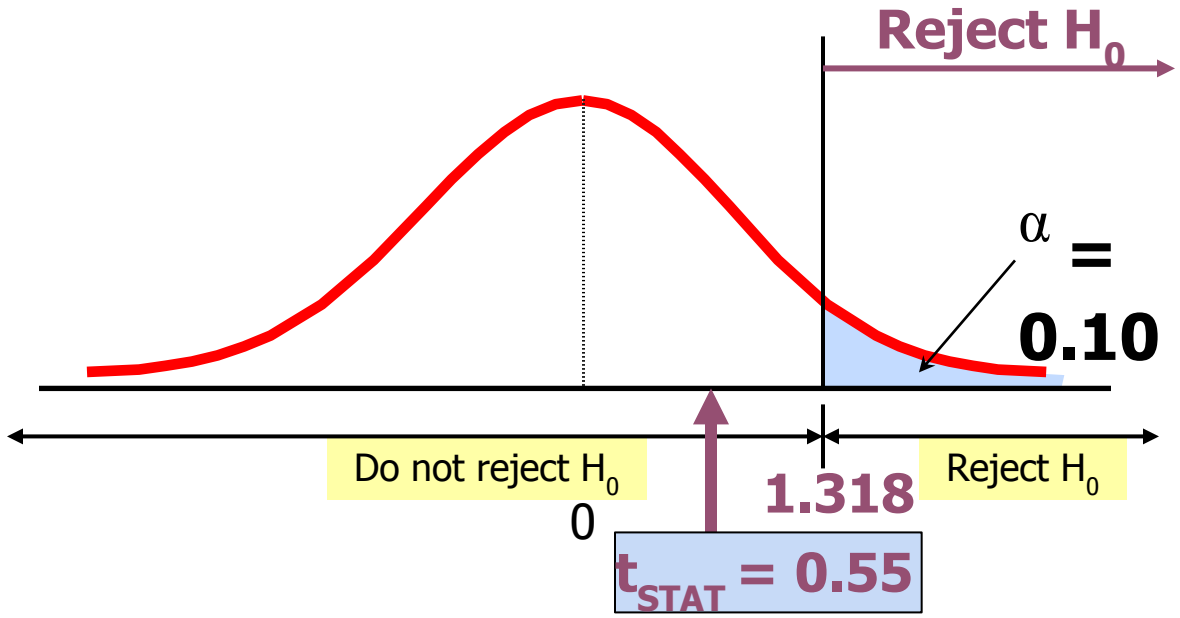
n = 25, $\bar{X}$ = 53.1, and S = 10.

Then the test statistic is:

$$t_{STAT} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{53.1 - 52}{\frac{10}{\sqrt{25}}} = 0.55$$

Example: Decision

Reach a decision and interpret the result.

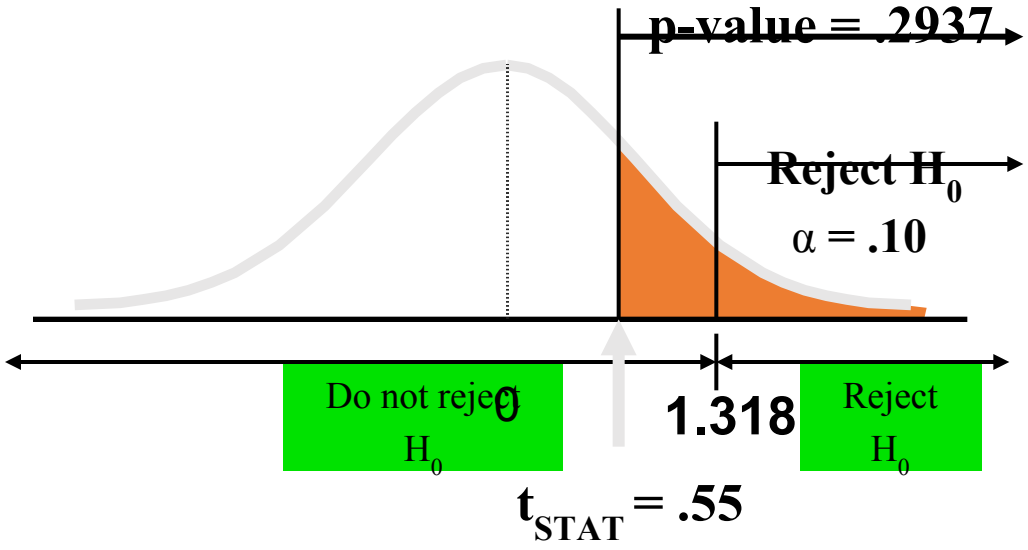


Do not reject H_0 since $t_{STAT} = 0.55 < 1.318$.

There is not sufficient evidence that the mean bill is over $52.

Example: Utilizing The p-value for The Upper Tail t-Test

- Calculate the p-value and compare to α (p-value below calculated using Excel spreadsheet on next page)



Do not reject H_0 since p-value = .2937 > α = .10

Excel Spreadsheet

Calculating The p-value for The Upper Tail t Test

t Test for the Hypothesis of the Mean		
Data		
Null Hypothesis μ=	**52.00**	
Level of Significance	**0.1**	
Sample Size	**25**	
Sample Mean	**53.10**	
Sample Standard Deviation	**10.00**	
Intermediate Calculations		
Standard Error of the Mean	2.00	=B8/SQRT(B6)
Degrees of Freedom	24	=B6-1
t test statistic	**0.55**	=(B7-B4)/B11
Upper Tail Test		
Upper Critical Value	1.318	=T.INV.2T2*B5,B12)
p-value	0.2937	=T.DIST.RT(ABS(B13),B12,1)
Do Not Reject Null Hypothesis		=IF(B18<B5, "Reject null hypothesis",
		"Do not reject null hypothesis")

END OF PART 1

Week 11: Hypothesis Testing

Paul Andrew Gorgonio